

Board Name

Variant: Draft

2024-04-06

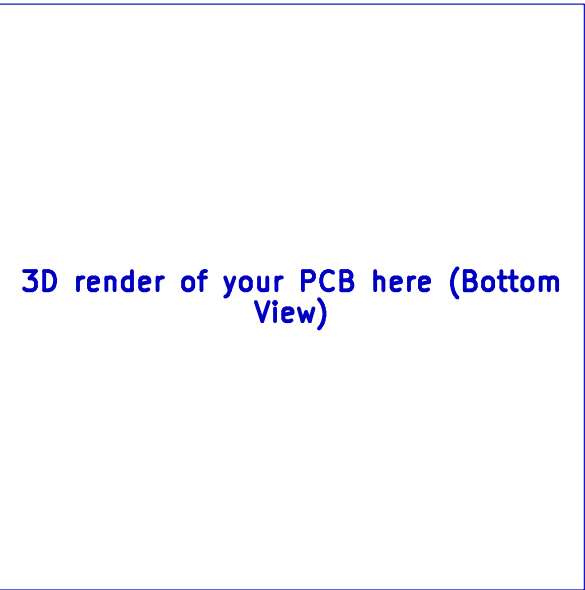
Rev 1.0

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TOP VIEW



BOTTOM VIEW



NOTES

Add a comment here

Not fitted components are marked as **X**

**DRAFT** - Very early stage of schematic, ignore details.  
**PRELIMINARY** - Close to final schematic.  
**CHECKED** - There shouldn't be any mistakes. Contact the engineer if you find any.  
**RELEASED** - A board with this schematic has been sent to production.

Draft Release Date

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

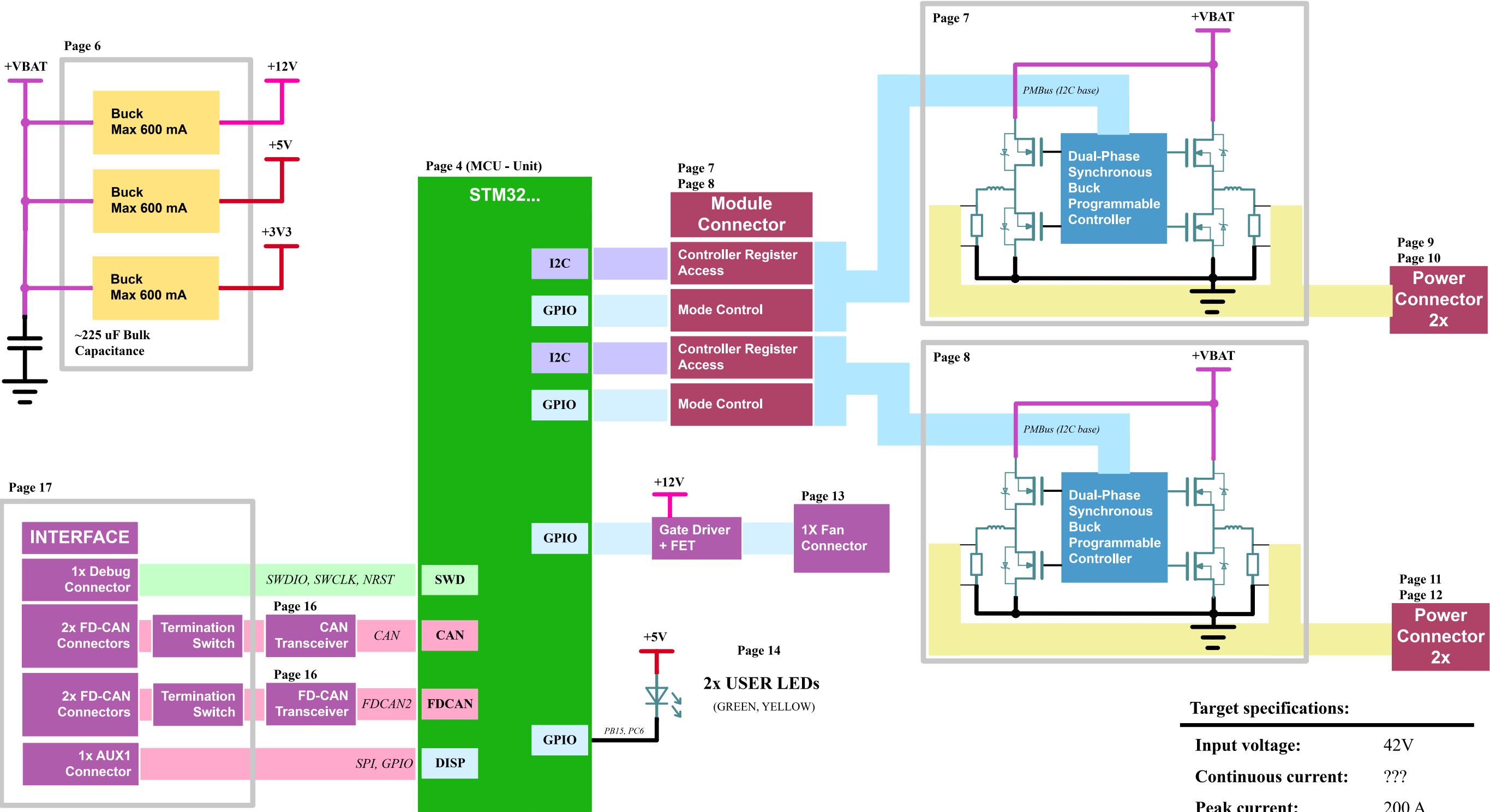
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

|  |                            |                                           |                            |                                    |                   |
|--|----------------------------|-------------------------------------------|----------------------------|------------------------------------|-------------------|
|  | Comments:                  | Company:<br>EPFL Xplore                   |                            | Variant:<br>Draft                  |                   |
|  | Sheet Title:<br>Cover Page | Board Name:<br>Board Name                 |                            | Project Name:<br>smps_legged_robot |                   |
|  | Sheet Path:<br>/           | File Name:<br>smps_legged_robot.kicad_sch | Designer:<br>Federico Bise | Date:<br>Last Modified Date        | Revision:<br>1.0  |
|  |                            |                                           | Reviewer:<br>Reviewer Name | Size:<br>A3                        | Sheet:<br>1 of 19 |

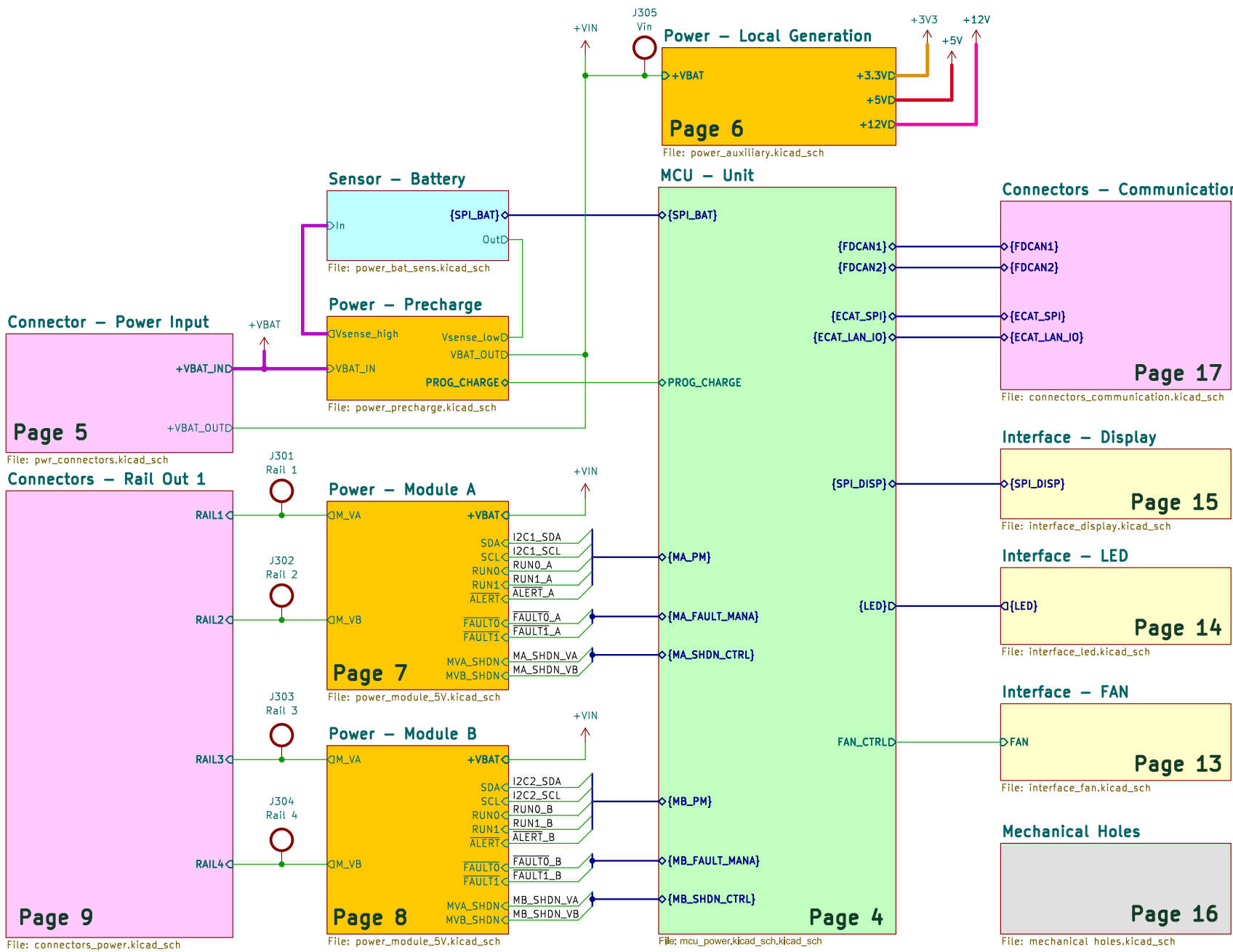
[2] Block Diagram



| Target specifications: |       |
|------------------------|-------|
| Input voltage:         | 42V   |
| Continuous current:    | ???   |
| Peak current:          | 200 A |
| Switching frequency:   | ???   |

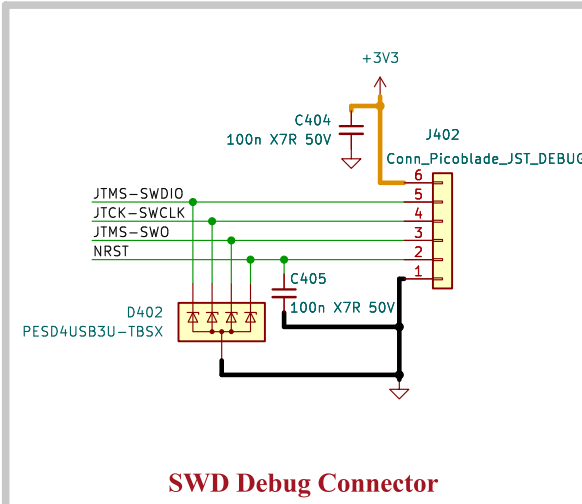
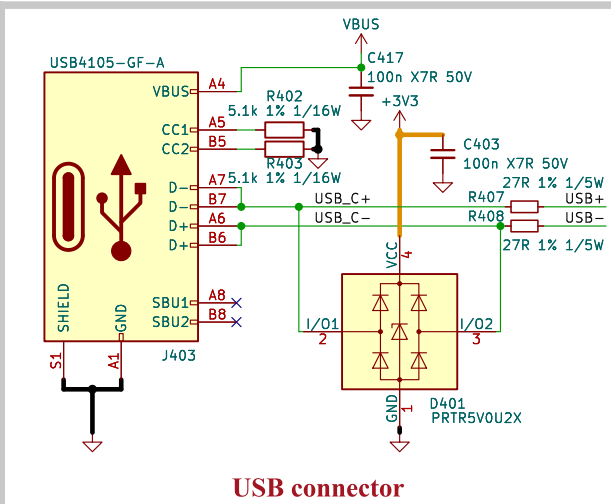
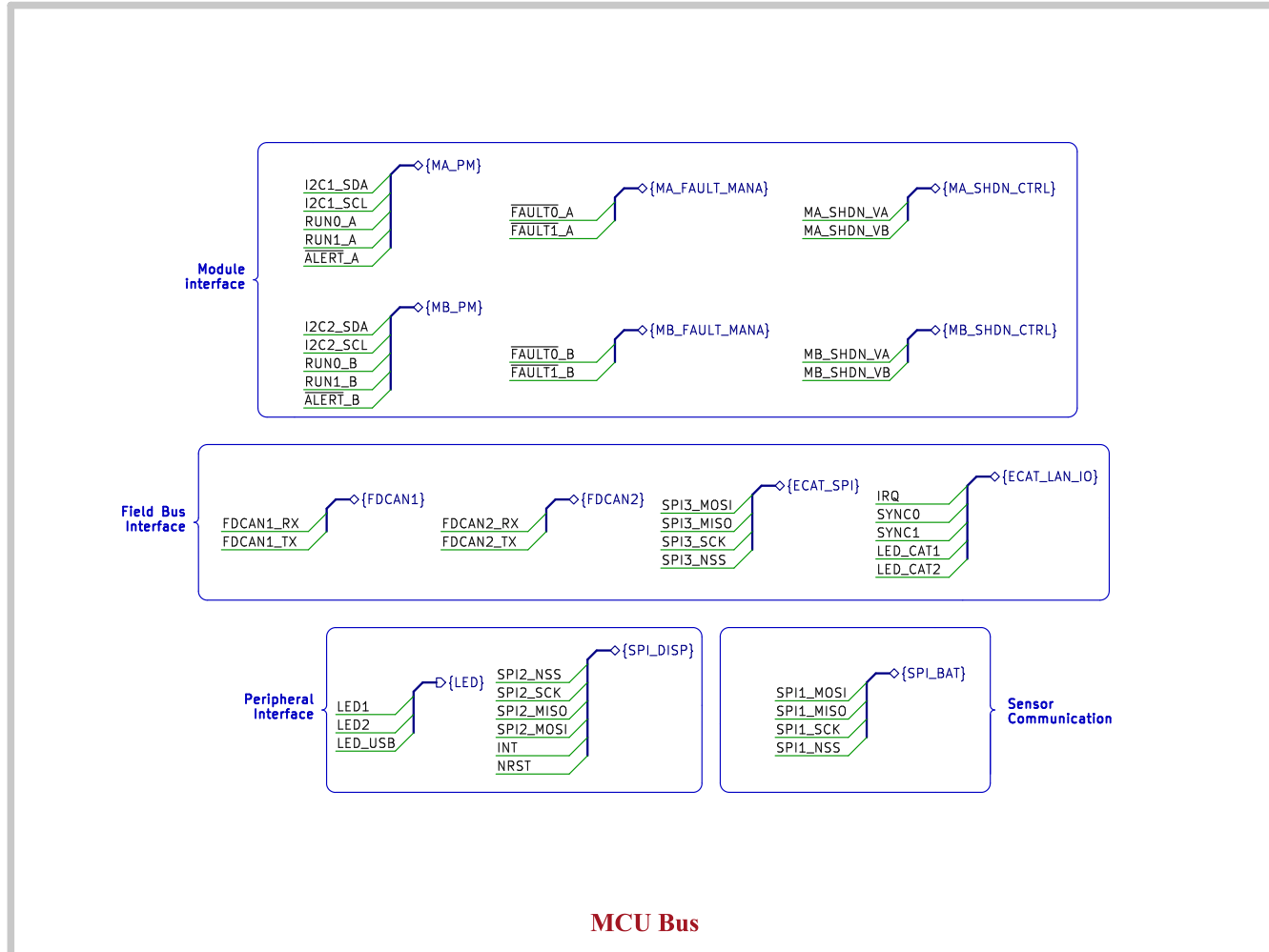
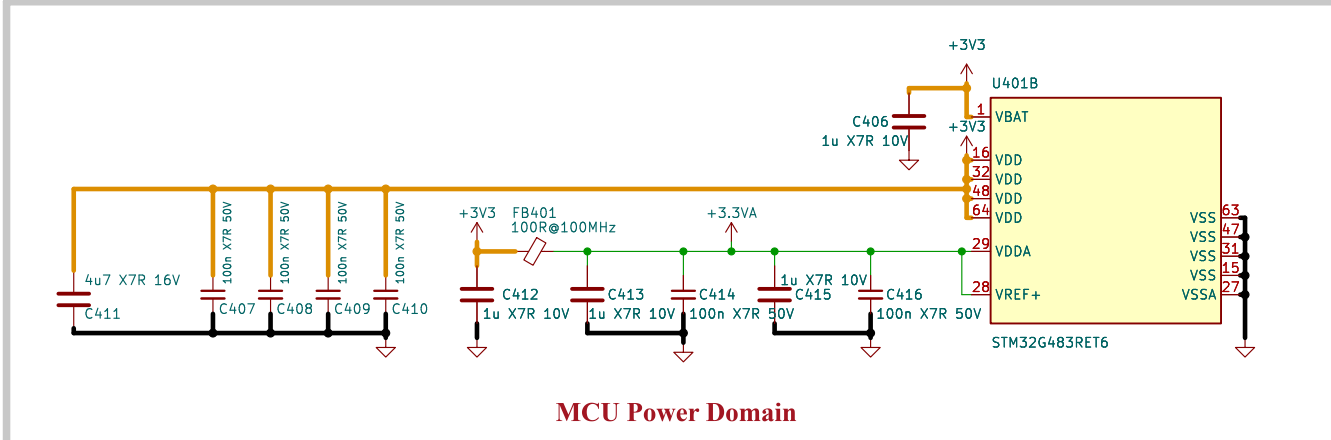
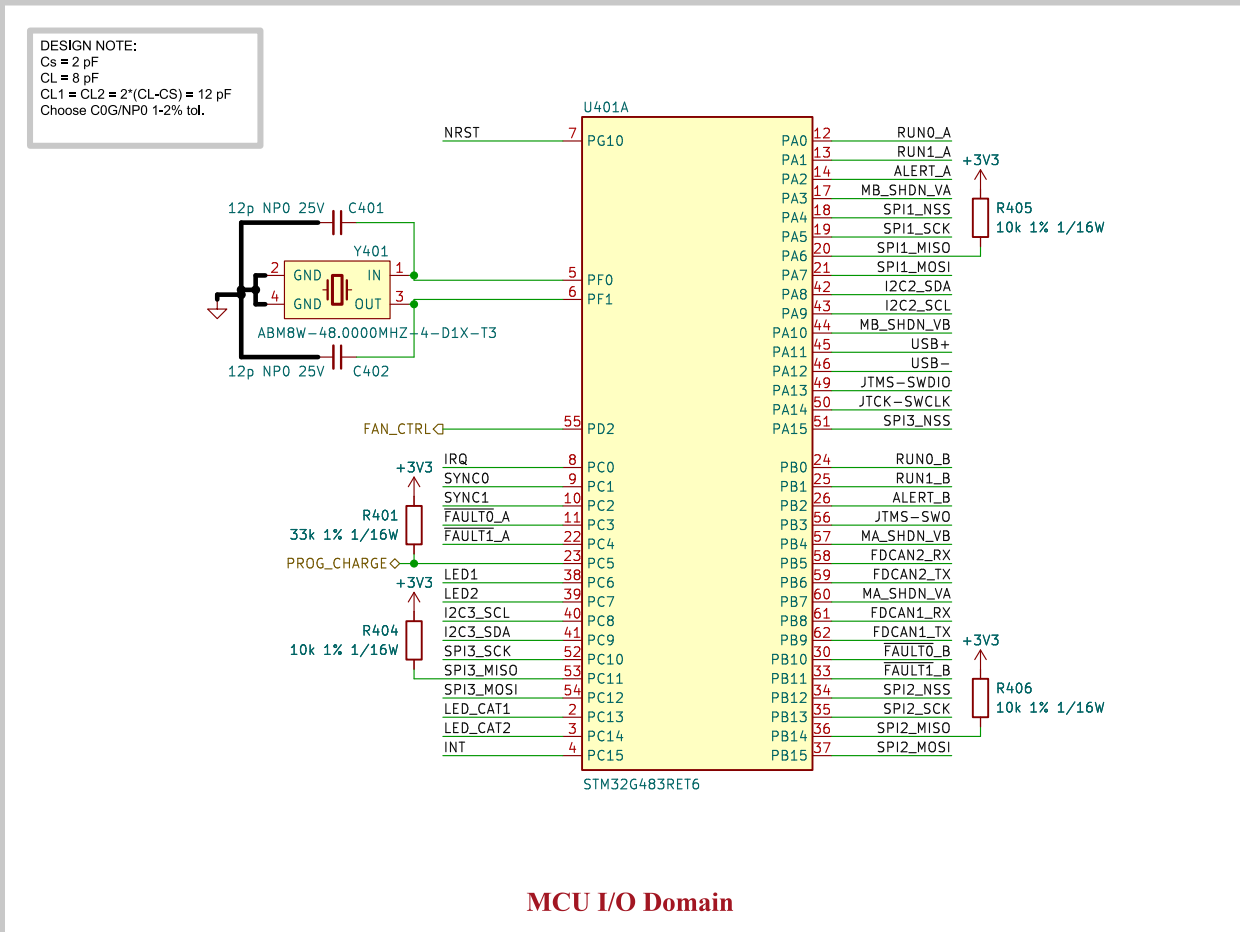
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|                                | Board Name:<br>Board Name     |                                       | Project Name:<br>smps_legged_robot |                             |
|                                | Sheet Title:<br>Block Diagram | File Name:<br>Block Diagram.kicad_sch | Designer:<br>Federico Bise         | Date:<br>Last Modified Date |
| Sheet Path:<br>/Block Diagram/ |                               | Reviewer:<br>Reviewer Name            | Size:<br>A3                        | Sheet:<br>2 of 19           |

# [3] Project Architecture



|                                       |                                      |                                              |                            |                             |                    |  |
|---------------------------------------|--------------------------------------|----------------------------------------------|----------------------------|-----------------------------|--------------------|--|
|                                       | Comments:                            |                                              | Company:<br>EPFL Xplore    |                             | Variant:<br>Draft  |  |
|                                       | Board Name:                          |                                              | Project Name:              |                             | smpls_legged_robot |  |
|                                       | Sheet Title:<br>Project Architecture | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Federico Bise | Date:<br>Last Modified Date | Revision:<br>1.0   |  |
| Sheet Path:<br>/Project Architecture/ |                                      |                                              | Reviewer:<br>Reviewer Name | Size:<br>A3                 | Sheet:<br>3 of 19  |  |

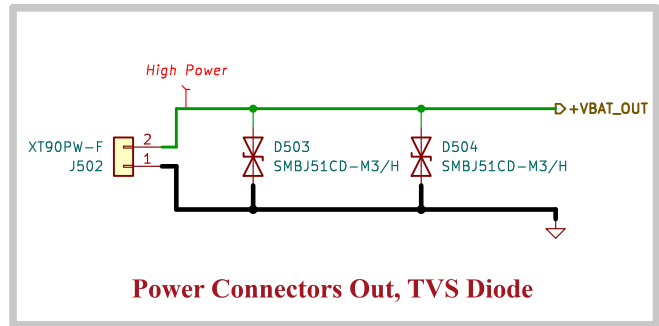
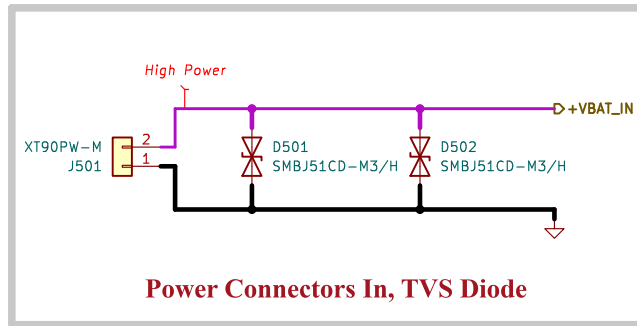
## [4] MCU - Unit



|  |                                                  |  |                                             |                            |                                           |                                 |
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|  | Comments:                                        |  | Company:<br><b>EPFL Xplore</b>              |                            | Variant:<br>Draft                         |                                 |
|  |                                                  |  | Board Name:<br><b>Board Name</b>            |                            | Project Name:<br><b>smps_legged_robot</b> |                                 |
|  | Sheet Title:<br>Sheet Title A                    |  | File Name:<br>mcu_power.kicad_sch.kicad_sch | Designer:<br>Federico Bise | Date:<br>Last Modified Date               | Revision:<br>1.0                |
|  | Sheet Path:<br>/Project Architecture/MCU - Unit/ |  |                                             | Reviewer:<br>Reviewer Name | Size:<br><b>A3</b>                        | Sheet:<br><b>4</b> of <b>19</b> |

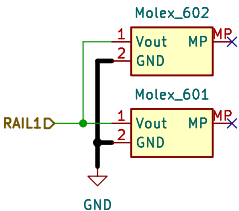


# [5] Connector - Power

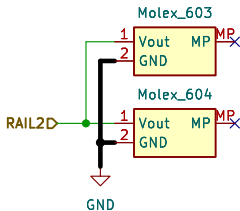


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|  |                                                |                          |               |       | Draft             |  |
|  |                                                |                          | Board Name:   |       | Project Name:     |  |
|  |                                                |                          | Board Name    |       | smps_legged_robot |  |
|  | Sheet Title:                                   | File Name:               | Designer:     | Date: | Revision:         |  |
|  |                                                | pwr_connectors.kicad_sch | Federico Bise |       |                   |  |
|  | Sheet Path:                                    |                          | Reviewer:     | Size: | Sheet:            |  |
|  | /Project Architecture/Connector - Power Input/ |                          | Reviewer Name | A5    | 5 of 19           |  |

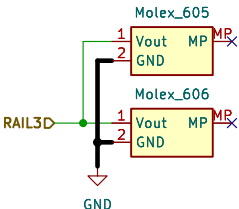
[9-12] Power - Power Out



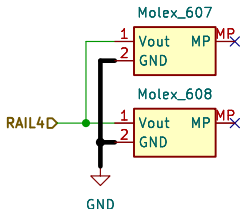
Output Rail 1



Output Rail 2



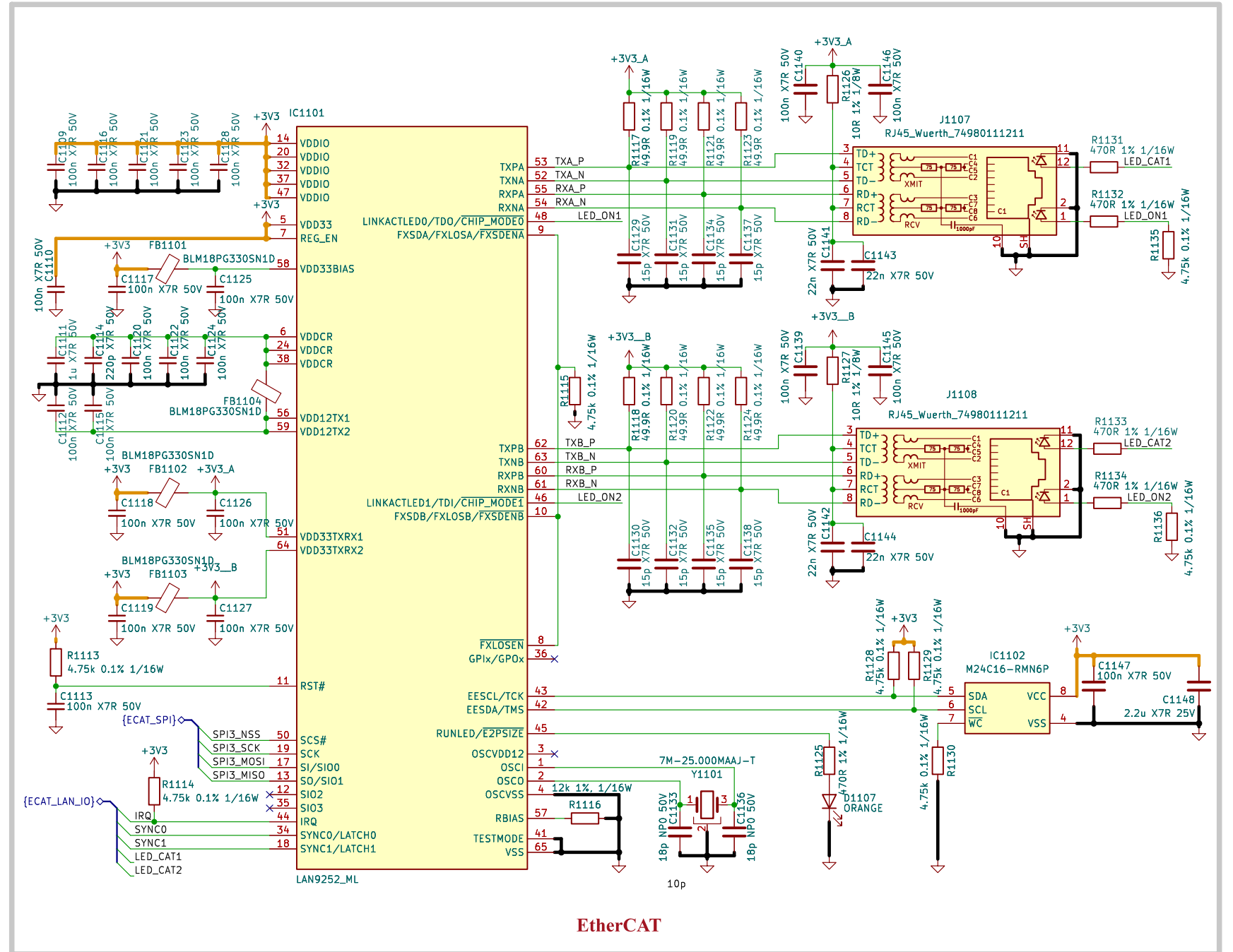
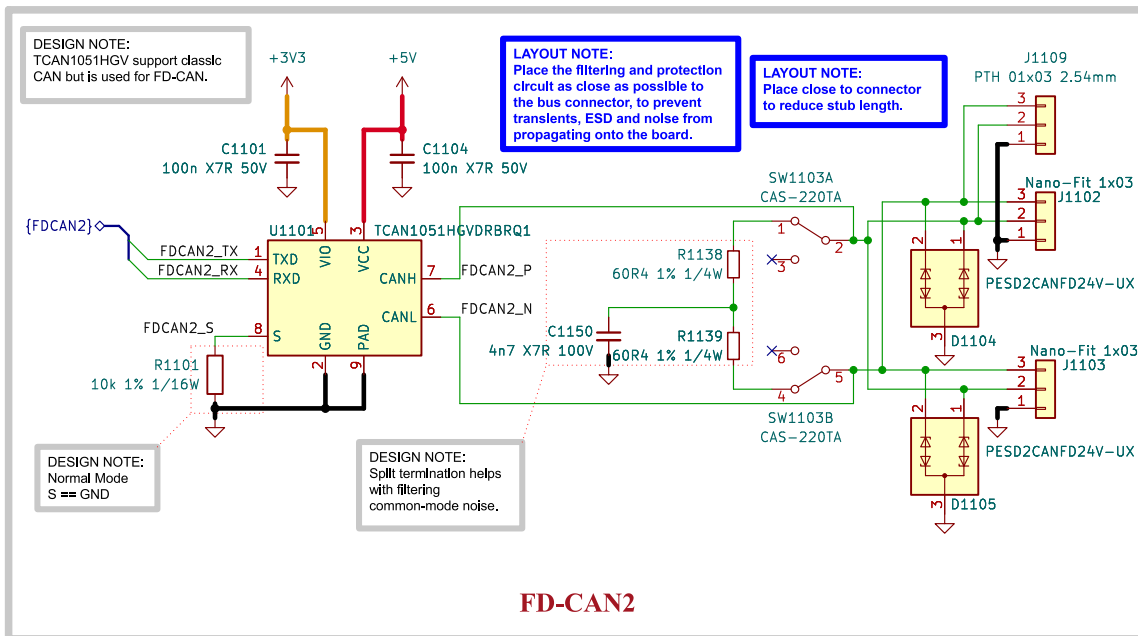
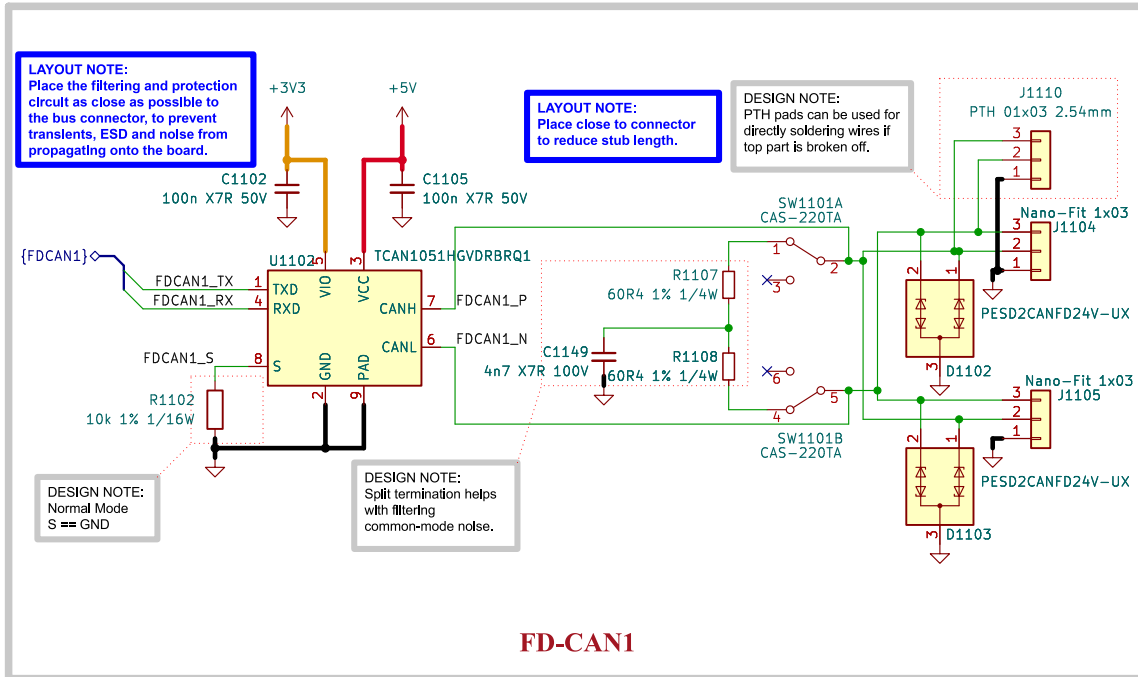
Output Rail 3



Output Rail 4

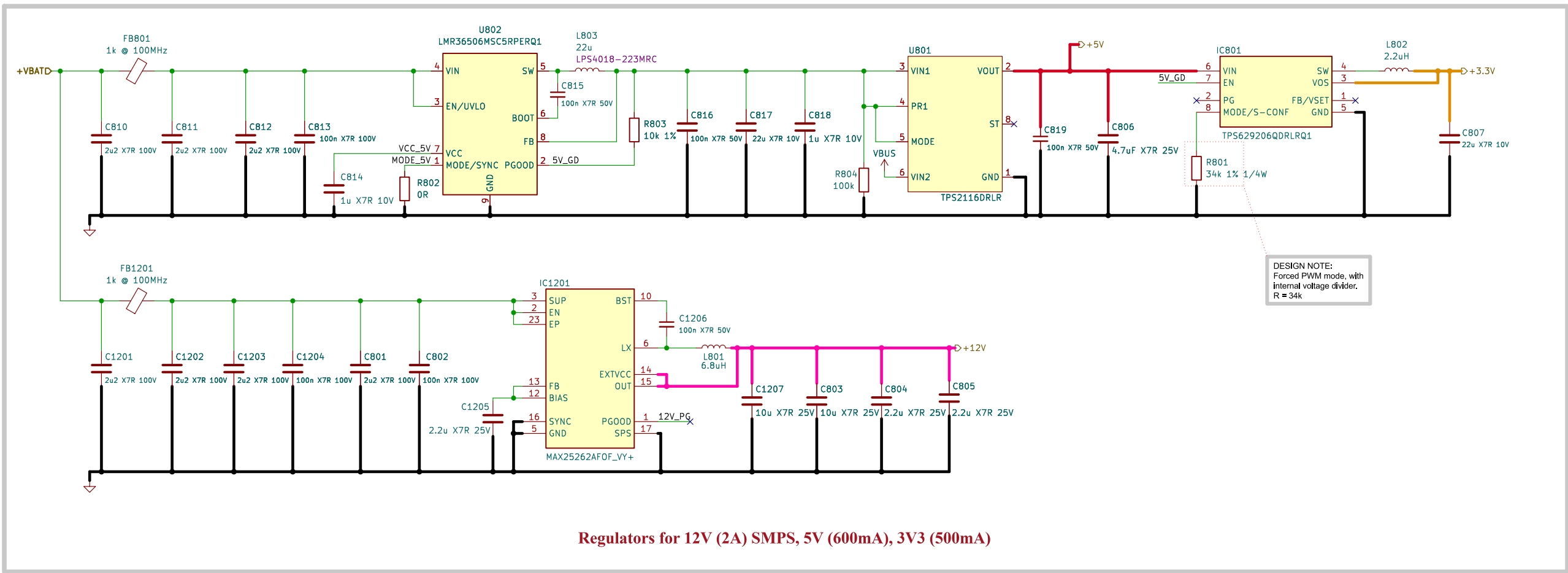
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|  |                                                | Board Name:                |               | Project Name:     |           |
|  |                                                | Board Name                 |               | smps_legged_robot |           |
|  | Sheet Title:                                   | File Name:                 | Designer:     | Date:             | Revision: |
|  |                                                | connectors_power.kicad_sch | Federico Bise |                   |           |
|  | Sheet Path:                                    |                            | Reviewer:     | Size:             | Sheet:    |
|  | /Project Architecture/Connectors - Rail Out 1/ |                            | Reviewer Name | A4                | 7 of 19   |

## [11] Interface - Communication



|                                                                  |  |                                                  |                            |                                           |                                  |
|------------------------------------------------------------------|--|--------------------------------------------------|----------------------------|-------------------------------------------|----------------------------------|
| Comments:                                                        |  | Company:                                         |                            | Variant:<br>Draft                         |                                  |
|                                                                  |  | Board Name:<br><b>Board Name</b>                 |                            | Project Name:<br><b>smps_legged_robot</b> |                                  |
| Sheet Title:                                                     |  | File Name:<br>connectors_communication.kicad_sch | Designer:<br>Federico Bise | Date:                                     | Revision:                        |
| Sheet Path:<br>/Project Architecture/Connectors - Communication/ |  |                                                  | Reviewer:<br>Reviewer Name | Size:<br><b>A3</b>                        | Sheet:<br><b>11</b> of <b>19</b> |

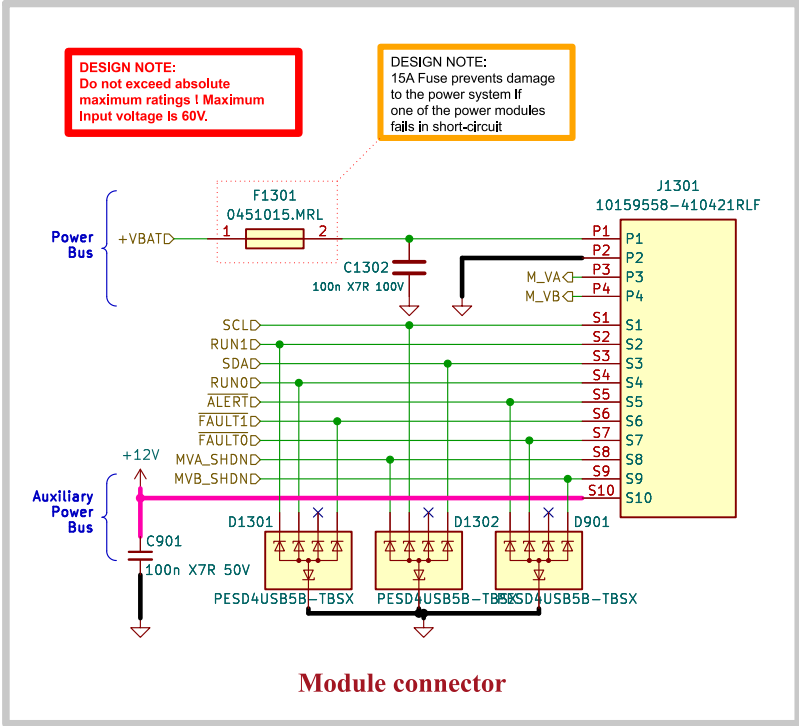
[12] Power - Local Generation



Regulators for 12V (2A) SMPS, 5V (600mA), 3V3 (500mA)

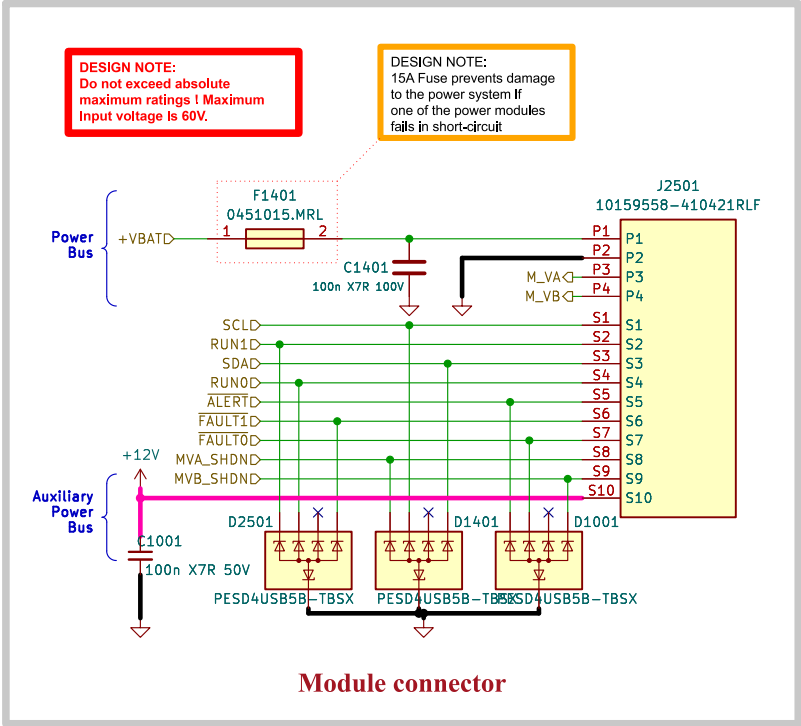
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|                                                 | Sheet Title: |                           | Board Name:   |       | Project Name:     |  |
|                                                 |              |                           | Board Name    |       | smps_legged_robot |  |
|                                                 |              | File Name:                | Designer:     | Date: | Revision:         |  |
|                                                 |              | power_auxiliary.kicad_sch | Federico Bise |       |                   |  |
| Sheet Path:                                     |              |                           | Reviewer:     | Size: | Sheet:            |  |
| /Project Architecture/Power - Local Generation/ |              |                           | Reviewer Name | A3    | 12 of 19          |  |

# [13-14] Power - Module



|  |                                         |  |                           |               |                   |           |
|--|-----------------------------------------|--|---------------------------|---------------|-------------------|-----------|
|  | Comments:                               |  | Company:                  |               | Variant:          |           |
|  |                                         |  |                           |               | Draft             |           |
|  |                                         |  | Board Name:               |               | Project Name:     |           |
|  |                                         |  | Board Name                |               | smgs_legged_robot |           |
|  | Sheet Title:                            |  | File Name:                | Designer:     | Date:             | Revision: |
|  |                                         |  | power_module_5V.kicad_sch | Federico Bise |                   |           |
|  | Sheet Path:                             |  |                           | Reviewer:     | Size:             | Sheet:    |
|  | /Project Architecture/Power - Module A/ |  |                           | Reviewer Name | A4                | 13 of 19  |

[13-14] Power - Module

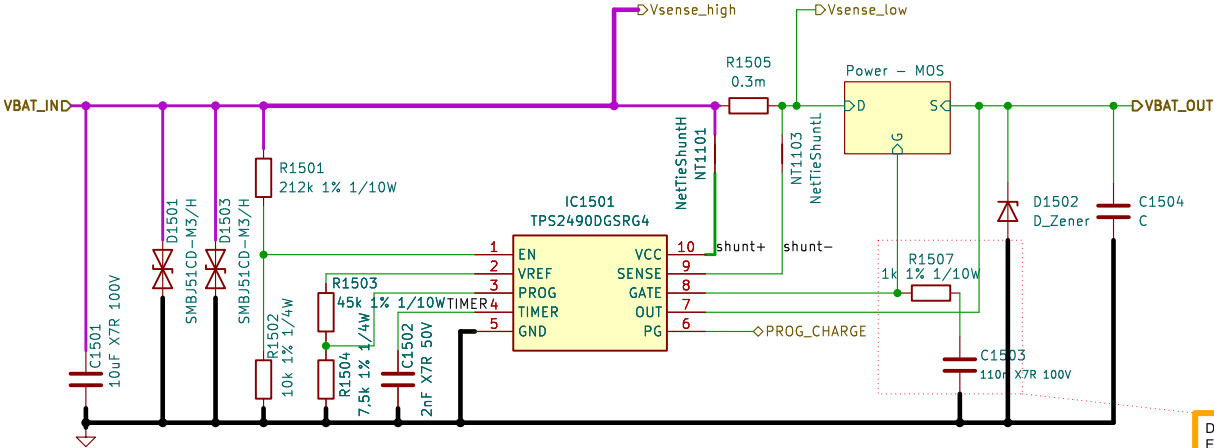


|  |                                                        |  |                                         |                            |                                           |                                  |
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|  | Comments:                                              |  | Company:                                |                            | Variant:<br>Draft                         |                                  |
|  |                                                        |  | Board Name:<br><b>Board Name</b>        |                            | Project Name:<br><b>smps_legged_robot</b> |                                  |
|  | Sheet Title:                                           |  | File Name:<br>power_module_5V.kicad_sch | Designer:<br>Federico Bise | Date:                                     | Revision:                        |
|  | Sheet Path:<br>/Project Architecture/Power - Module B/ |  |                                         | Reviewer:<br>Reviewer Name | Size:<br><b>A4</b>                        | Sheet:<br><b>14</b> of <b>19</b> |

# [15]Power - Precharge

DESIGN NOTE:  
Max 6 parallel  
MOS.  
Up to 10000uF  
bulk capacitance

DESIGN NOTE:  
Max Current Limit :  
183,1A  
Max Nominal  
Current : 150A



DESIGN NOTE:  
External soft start up  
control

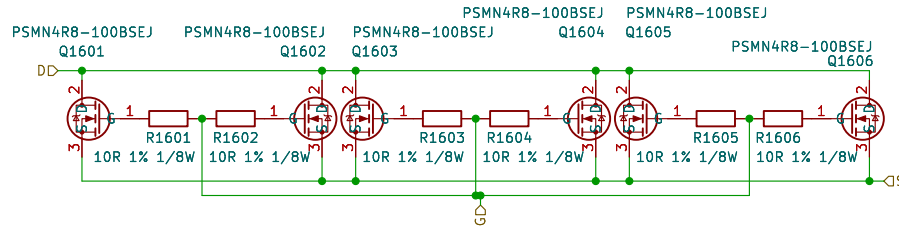
Power Hot-Swap Circuit

|  |                                          |                           |               |                    |           |
|--|------------------------------------------|---------------------------|---------------|--------------------|-----------|
|  | Comments:                                | Company:                  |               | Variant:           |           |
|  |                                          |                           |               | Draft              |           |
|  | Sheet Title:                             | Board Name:               |               | Project Name:      |           |
|  |                                          | Board Name                |               | smpls_legged_robot |           |
|  | Sheet Path:                              | File Name:                | Designer:     | Date:              | Revision: |
|  | /Project Architecture/Power - Precharge/ | power_precharge.kicad_sch | Federico Bise |                    |           |
|  | Sheet Path:                              | Reviewer:                 | Size:         | Sheet:             |           |
|  |                                          | Reviewer Name             | A4            | 15 of 19           |           |

# [16] Power - MOS

**DESIGN NOTE:**  
Max 6 parallel  
MOS.  
Up to 10000uF  
bulk capacitance

**DESIGN NOTE:**  
Maximum steady  
state FET Junction  
Temperature :  
129.1 °C



**Power Hot-Swap Circuit**

Comments:

Company:

Variant:

Draft

Board Name:

**Board Name**

Project Name:

**smps\_legged\_robot**

Sheet Title:

File Name:

power\_mos.kicad\_sch

Designer:

Federico Bise

Date:

Revision:

Sheet Path:

/Project Architecture/Power - Precharge/Power - MOS/

Reviewer:

Reviewer Name

Size:

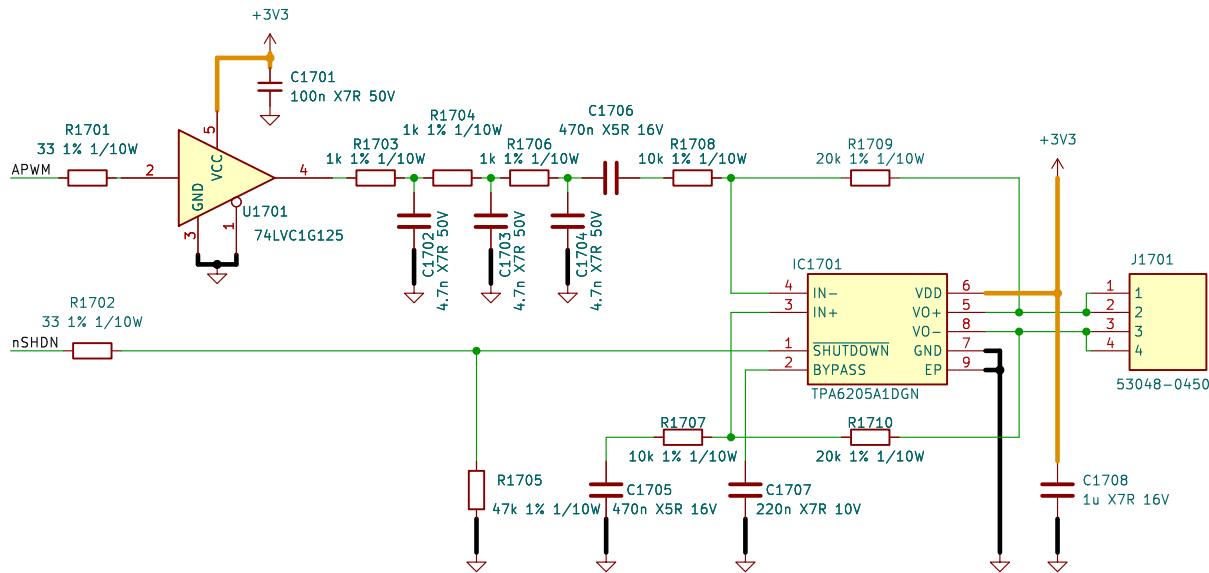
**A5**

Sheet:

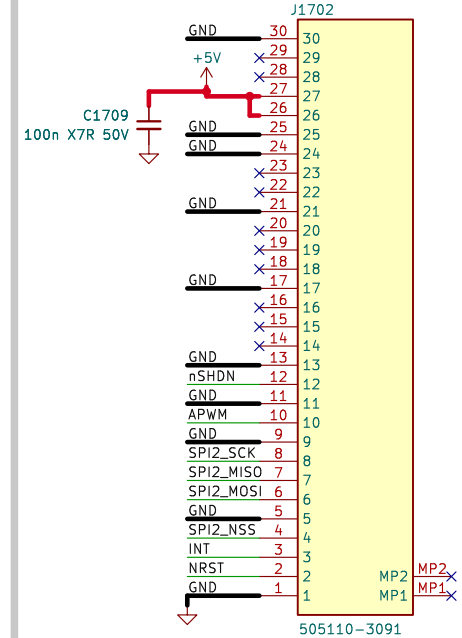
**16** of **19**



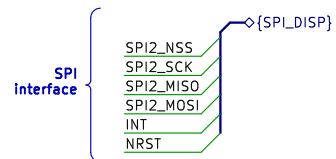
# [17] Interface - Display



Audio amplifier



External Connector



Comments:

Company:

Variant:

Draft

Board Name:

Board Name

Project Name:

smps\_legged\_robot

Sheet Title:

File Name:

interface\_display.kicad\_sch

Designer:

Federico Bise

Date:

Revision:

Sheet Path:

/Project Architecture/Interface - Display/

Reviewer:

Reviewer Name

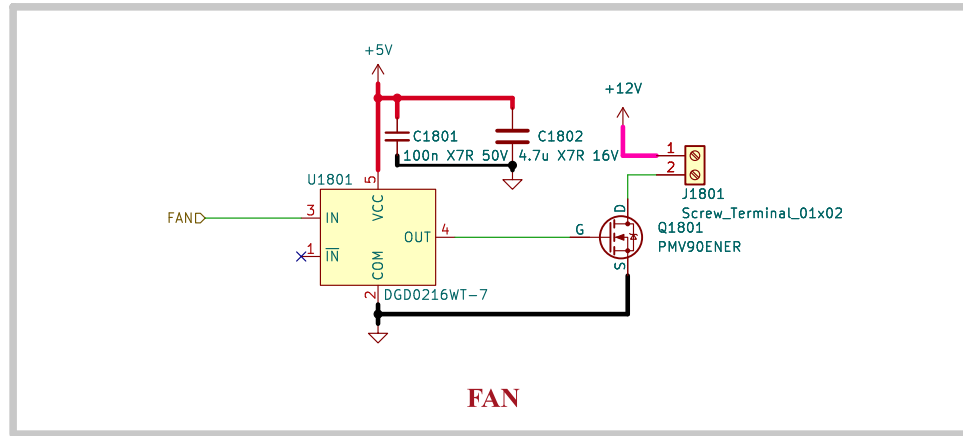
Size:

A4

Sheet:

17 of 19

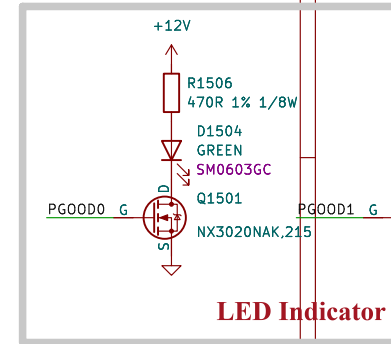
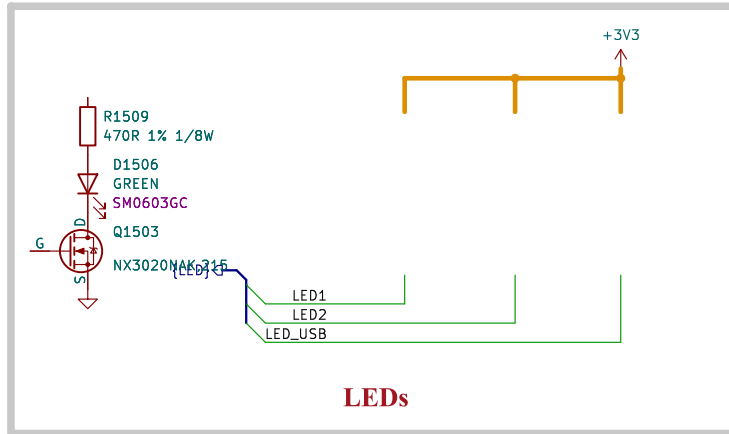
# [18] Interface - FAN



FAN

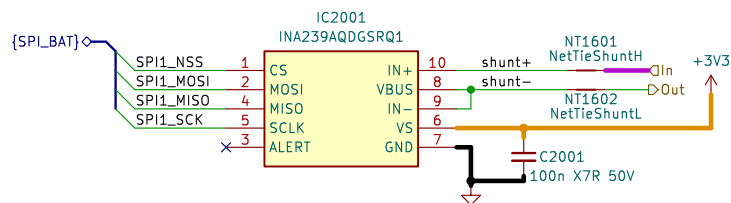
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|                                         |              |                         | Board Name:   |       | Project Name:      |  |
|                                         |              |                         | Board Name    |       | smmps_legged_robot |  |
|                                         | Sheet Title: | File Name:              | Designer:     | Date: | Revision:          |  |
|                                         |              | interface_fan.kicad_sch | Federico Bise |       |                    |  |
| Sheet Path:                             |              |                         | Reviewer:     | Size: | Sheet:             |  |
| /Project Architecture/Interface - FAN / |              |                         | Reviewer Name | A5    | 18 of 19           |  |

# [19] Interface - LED



|  |                                        |                         |                   |           |                          |  |
|--|----------------------------------------|-------------------------|-------------------|-----------|--------------------------|--|
|  | Comments:                              |                         | Company:          |           | Variant:                 |  |
|  |                                        |                         |                   |           | Draft                    |  |
|  |                                        |                         | Board Name:       |           | Project Name:            |  |
|  |                                        |                         | <b>Board Name</b> |           | <b>smps_legged_robot</b> |  |
|  | Sheet Title:                           | File Name:              | Designer:         | Date:     | Revision:                |  |
|  |                                        | interface_led.kicad_sch | Federico Bise     |           |                          |  |
|  | Sheet Path:                            |                         | Reviewer:         | Size:     | Sheet:                   |  |
|  | /Project Architecture/Interface - LED/ |                         | Reviewer Name     | <b>A5</b> | <b>19</b> of <b>19</b>   |  |

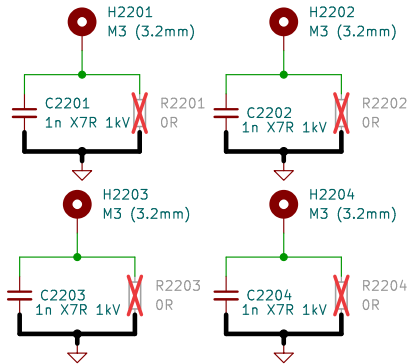
# [20-21] Sensor - Batterie



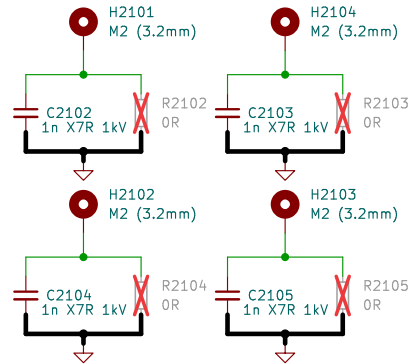
**Batterie Voltage, Current and Temperature Sensor**

|  |                                         |                          |                   |           |                          |  |
|--|-----------------------------------------|--------------------------|-------------------|-----------|--------------------------|--|
|  | Comments:                               |                          | Company:          |           | Variant:                 |  |
|  |                                         |                          |                   |           | Draft                    |  |
|  |                                         |                          | Board Name:       |           | Project Name:            |  |
|  |                                         |                          | <b>Board Name</b> |           | <b>smps_legged_robot</b> |  |
|  | Sheet Title:                            | File Name:               | Designer:         | Date:     | Revision:                |  |
|  |                                         | power_bat_sens.kicad_sch | Federico Bise     |           |                          |  |
|  | Sheet Path:                             |                          | Reviewer:         | Size:     | Sheet:                   |  |
|  | /Project Architecture/Sensor - Battery/ |                          | Reviewer Name     | <b>A5</b> | <b>20</b> of <b>19</b>   |  |

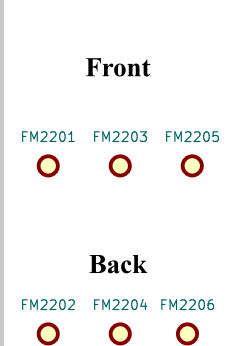
# [22] Mechanical Holes



**Drilling Holes**



**Module Holes**



**Fiducial Markers**

Comments:

Company:

Variant:

Draft

Board Name:

**Board Name**

Project Name:

**smps\_legged\_robot**

Sheet Title:

File Name:

mechanical\_holes.kicad\_sch

Designer:

Federico Bise

Date:

Revision:

Sheet Path:

/Project Architecture/Mechanical Holes/

Reviewer:

Reviewer Name

Size:

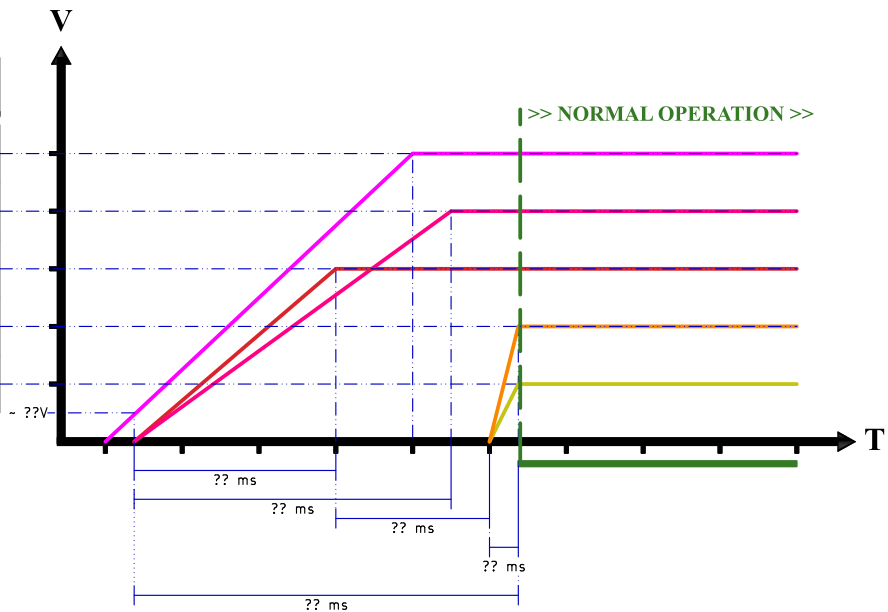
**A5**

Sheet:

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[23] Power - Sequencing

| NAME | SOURCE      | LEVEL     |
|------|-------------|-----------|
| +??V | SOURCE NAME | ??V ± ??% |
| +??V | SOURCE NAME | ??V ± ??% |
| +??V | SOURCE NAME | ??V ± ??% |
| +??V | SOURCE NAME | ??V ± ??% |
| +??V | SOURCE NAME | ??V ± ??% |



|  |                                     |                                            |                            |                                    |                    |
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|  | Comments:                           | Company:<br>EPFL Xplore                    |                            | Variant:<br>Draft                  |                    |
|  |                                     | Board Name:<br>Board Name                  |                            | Project Name:<br>smps_legged_robot |                    |
|  | Sheet Title:<br>Power - Sequencing  | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>Federico Bise | Date:<br>Last Modified Date        | Revision:<br>1.0   |
|  | Sheet Path:<br>/Power - Sequencing/ |                                            | Reviewer:<br>Reviewer Name | Size:<br>A4                        | Sheet:<br>23 of 19 |

# [24] Revision History

DD.MM.YYYY - xxx Revision  
Variant: xxx

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DD.MM.YYYY - xxx Revision  
Variant: xxx

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DD.MM.YYYY - xxx Revision  
Variant: xxx

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|  | Comments:                         |  | Company:<br>EPFL Xplore                  |                            | Variant:<br>Draft                  |                    |
|  |                                   |  | Board Name:<br>Board Name                |                            | Project Name:<br>smps_legged_robot |                    |
|  | Sheet Title:<br>Revision History  |  | File Name:<br>Revision History.kicad_sch | Designer:<br>Federico Bise | Date:<br>Last Modified Date        | Revision:<br>1.0   |
|  | Sheet Path:<br>/Revision History/ |  |                                          | Reviewer:<br>Reviewer Name | Size:<br>A4                        | Sheet:<br>24 of 19 |